GALGOTIAS UNIVERSITY

ADMISSION NUMBER

[redacted]

# School of Computer Applications and Technology

**Mid Term Examination - Apr 2026**

**M.Sc. CS II Sem / M.Sc. DS II Sem / MCA (All) II Sem / Int. BCA+MCA VIII Sem**

Duration : 90 Minutes
Max Marks : 50

## Sem II - E1PY203B - Data Structures

*General Instructions*
*Answer to the specific question asked*
*Draw neat, labelled diagrams wherever necessary*
*Approved data hand books are allowed subject to verification by the Invigilator*

| | | |
|---|---|---|
| 1) | What is a Sparse Matrix? Explain its representation. | CO1(3) |
| 2) | Explain the concept of Time-Space Trade-off with a suitable example. | CO2(5) |
| 3) | Explain Single Dimensional and Multidimensional Arrays with examples. | CO3(5) |
| 4) | Write a recursive function for Tower of Hanoi and explain: Number of moves (for n= 3) | CO4(8) |
| 5) | What is a dequeue (double ended queue) and explain its types such as input restricted dequeue and output restricted dequeue. | CO5(8) |
| 6) | Explain the applications of queues in real life and computer systems with suitable examples. | CO6(9) |
| 7) | Describe the structure and working of a doubly linked list and explain its advantages over a singly linked list with necessary operations. | CO5(12) |
| | **OR** | |
| | A photo viewer application allows users to move both forward and backward through images, also design a solution using a doubly linked list and explain its operations. | CO6(12) |